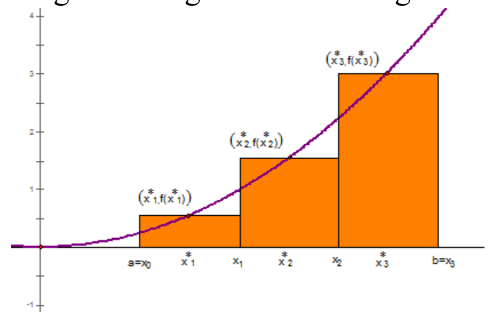
5.2.1 The Definite Integral

In the last section we discussed how we could approximate areas of regions using areas of rectangles.



Definition: The net area (also called signed area) of a region that lies between the x-axis and the graph of a continuous function f may be approximated with a <u>Riemann sum</u> which is a sum of the form

$$\sum_{i=1}^{n} f\left(x_i^*\right)\Delta x = f\left(x_1^*\right)\Delta x + f\left(x_2^*\right)\Delta x + f\left(x_3^*\right)\Delta x + \cdots + f\left(x_n^*\right)\Delta x$$

where x_i^* is a sample point in $\left[x_{i-1}, x_i\right]$ and $\Delta x = \dfrac{b-a}{n}$.

<u>Example 1:</u> Consider the area between the function $f(x) = x^2$ and the x-axis on the interval $[0,2]$. We will approximate this area using a right endpoint estimate with 4 rectangles.

a) Carefully draw and label a graph representing this approximation.

b) Express this approximation for the area using sigma notation.

Question: When it comes to Riemann sums what do we need to modify in order to improve our approximation? Could we use a Riemann sum to obtain the actual value of the net area?

Definition: Suppose that $\sum_{i=1}^{n} f\left(x_i^*\right)\Delta x$ is a Riemann sum which approximates the net area A of a region that lies between a continuous function f and the x-axis. Then,

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right)\Delta x$$

Note: It can be shown in higher level math courses (namely real analysis) that the limit in the above definition will be the same regardless of which sample points are used. For simplicity, we will usually choose to do a left endpoint or right endpoint sum when applying this definition.

Example 2: Let A be the area of the region that lies under the graph and above the x-axis of $f(x) = x^2$ between $x=0$ & $x=2$. Using right endpoints, find an expression for A as a limit.

The next logical question to ask would be "How can we evaluate a limit like this?"
At times it is surprising easy to compute such limits if we are aware of the following facts:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

These facts can be shown using the idea of *Mathematical Induction* which is proof technique often discussed in Precalculus.

Example 3: Determine the *exact* area between $f(x) = x^2$ and the *x*-axis on the interval $[0,2]$.

Definition: Suppose f is a function defined on $[a,b]$. Suppose we divide $[a,b]$ into *n* subintervals of width Δx. We let $a = x_0$, x_1, x_2, …, and $b = x_n$ be the endpoints of these subintervals, and we let x_1^*, x_2^*, …, and x_n^* be sample points in each of the subintervals.

Then, <u>the definite integral of f from *a* to *b*</u> is:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points.
If it does exist, we say that f is <u>integrable</u> on $[a,b]$.

Note: We should read $\int_a^b f(x)\,dx$ *as one large symbol which indicates a net area.*

Note: This definition allows us to choose <u>any</u> sample point in each subinterval. We will often choose the right endpoint for simplicity.

Terminology:

- The symbol $\int$ is the <u>integral sign</u>.
- For $\int_a^b f(x)\,dx$, $f(x)$ is called the <u>integrand</u>.
- For $\int_a^b f(x)\,dx$, a and b are called the <u>limits of integrations</u>. Specifically, a is the <u>lower limit</u> and b is the <u>upper limit</u>.
- We think of dx as indicating the variable of integration.

<u>Example 4:</u> Consider $\int_{-1}^{1} x\,dx$. Draw a picture of the net area that this definite integral represents. Based upon your picture what would you suspect this definite integral is equal to?

Note: A definite integral $\int_a^b f(x)\,dx$ **<u>is a number</u>**; it does not depend on the variable of integration.

So, $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt = \int_a^b f(r)\,dr$.

One should notice that in our definition of definite integral we did not assume that the function of integration was continuous. The next theorem tells us that most functions we run into are in fact integrable.

Theorem: If f is continuous on $[a,b]$, or if f only has a finite number of jump discontinuities on $[a,b]$, then f is integrable on $[a,b]$; that is, the definite integral $\int_a^b f(x)\,dx$ exists.

Proof: Beyond the scope of this course

In order to evaluate definite integrals we need to know how to work with sums. The following formulas for sums will be needed.

(1) $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

(2) $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

(3) $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$

(4) $\sum_{i=1}^{n} c = nc$

(5) $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$

(6) $\sum_{i=1}^{n} (a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$

<u>Example 5:</u> Calculate the definite integral $\int_{-1}^{3} (x^2 - 2)\,dx$.

Example 6: Calculate the definite integral $\int_0^3 \left(x^3 - 6x\right) dx$.

Example 7: Let A be the area of the region that lies under the graph and above the x-axis of $f(x)=\sqrt{2-x}$ between $x=-5$ & $x=1$.

a) State the definite integral which is equal to A.

b) Using right endpoints, find an expression for A as a limit.